Introduction

2Market is a global supermarket that sells its products both online and through physical stores. The purpose of this study is to evaluate whether customer demographics, location and marketing channels have a direct impact on purchasing behaviour. These insights will support marketing executives in allocating budgets more effectively to drive global revenue growth.

The 5 Whys framework was employed to identify the root cause of the problem and establish the project objectives (Appendix 1); however, its simplicity limits the scope of the project and ability to explore the full depth and complexity of the issue.

The study focuses on assessing whether:

- customer demographics influence spending behaviour;
- different marketing approaches lead to successful customer conversions; and
- these patterns vary across geographic regions.

Two datasets were provided: one containing customer demographics and spending across product categories, and another detailing successful conversions by marketing channel per customer. As transactional data was unavailable, spending and conversions are assumed to represent cumulative totals for each customer.

Analysis

Prior to exploratory analysis in Excel, the dataset was cleaned to ensure accuracy and consistency. The data was first reviewed for blank or null values, with none identified. Field formats were validated and corrected where required, particularly for date and income fields. A total of 47 duplicate records were removed. Customer age was not explicitly provided and was derived by subtracting the year of birth from the current year (2025). Several marital status labels were standardised to improve data consistency: “Absurd” and “YOLO” were reclassified as “Unknown”; “Alone” was reclassified as “Single”; and “Married” and “Together” were consolidated into “Partnership”.

Three extreme age outliers were identified and deemed implausible (Table 1). These records were corrected by assigning the average year of birth (1970), giving an adjusted age of 55. One income outlier (Table 2) was similarly corrected by replacing it with the average income of $52,247 to preserve record completeness while improving reliability.

ID	Year of Birth	Age
11004	1894	131
1150	1900	125
7829	1901	124

Table 1: age outliers.

ID	Income
9432	$ 666,666

Table 2: income outlier.

Initial analysis shows that most customers are in a partnership, with an average age of 55, followed by single customers with an average age of 52 (Table 3). Single customers have the lowest average income ($50,856), while customers in partnerships earn slightly more on average ($51,942) (Table 4).

Marital Status	Average Age	Count
Single	52	464
Partnership	55	1399
Divorced	58	226
Widow	65	73
Unknown	50	4

Table 3: average age and customer count per marital status group.

Marital Status	Average Income
Divorced	$52,796
Partnership	$51,942
Single	$50,856
Unknown	$60,399
Widow	$56,707

Table 4: average income per marital status group.

The cleaned data was imported into pgAdmin for further analysis using SQL. Two tables were created from CSV files and queried to calculate customer counts by country and marital status, total sales by country and product category, and percentage distributions (see Appendix B for full syntax). Spain accounts for nearly half of all customers (48.61%), followed by South Africa (15.33%) and Canada (12.14%) (Table 5). Sales distribution mirrors customer concentration, with Spain generating 48.4% of total sales, followed by South Africa (15.66%) and Canada (12.71%) (Table 6). Customers in partnerships contribute the highest total spend due to their larger population share (Table C1, Appendix C). Across all countries, alcohol products are the top-selling category, followed by meat (Table C2, Appendix C).

	Country character varying (10)	number_of_customers bigint	pct_of_total numeric
1	SP	1053	48.61
2	SA	332	15.33
3	CA	263	12.14
4	AUS	146	6.74
5	IND	146	6.74
6	GER	116	5.36
7	US	107	4.94
8	ME	3	0.14

Table 5: total number of customers & their percentage grouped by country.

	Country character varying (10)	total_sales numeric	pct_of_total numeric
1	SP	636151	48.40
2	SA	205834	15.66
3	CA	167077	12.71
4	AUS	84970	6.46
5	IND	76572	5.83
6	GER	73198	5.57
7	US	67546	5.14
8	ME	3122	0.24

Table 6: total amount of sales ($) and their percentages grouped by country.

To analyse marketing performance, a table was created from the datasets using an inner join. Total conversions are highest in Spain (Table 7), reflecting its largest customer base and highest sales volume. Conversion performance across marketing channels differs insignificantly: Twitter, Instagram, and bulk email each contribute ~25% of conversions, Facebook accounts for just over 20%, only brochures perform significantly lower at ~5% (Table 8).

	Country character varying (10)	bulkmail bigint	twitter bigint	instagram bigint	facebook bigint	brochure bigint	conversions numeric	total_sales numeric
1	SP	80	85	87	74	15	341	636151
2	SA	21	20	21	20	4	86	205834
3	CA	17	24	21	18	6	86	167077
4	AUS	9	6	12	7	0	34	84970
5	IND	13	10	6	7	2	38	76572
6	GER	10	11	8	7	2	38	73198
7	US	8	6	5	7	0	26	67546
8	ME	1	0	0	0	0	1	3122

Table 7: conversions per marketing channel, total conversions and total sales grouped by country.

	channel text	total_conversions numeric	pct_of_total numeric
1	twitter	162	24.92
2	instagram	160	24.62
3	bulkmail	159	24.46
4	facebook	140	21.54
5	brochure	29	4.46

Table 8: total conversions per marketing channel and their percentages.

Dashboard

The new data table was exported into Tableau to enable final analysis and more effective visualisations. A Tableau Story was developed using three dashboards: one displaying customer demographics, one spending patterns, and one marketing channel performance. A consistent colour theme was applied using the Nuriel Stone palette, selected for its neutral tone, strong contrast, and avoidance of red–green combinations to support accessibility for colour-blind users. The dashboards include interactive filters which enable users to derive more granular insights, as required.

Each dashboard was limited to a maximum of four visualisations to maintain clarity, optimal layout, and readability. The demographics dashboard (Figure 1) presents key summary statistics alongside a pie chart illustrating customer distribution by country. A bar chart indicates that the largest proportion of customers fall within the 45–55 age group and that, across all age groups, most customers are in partnerships.

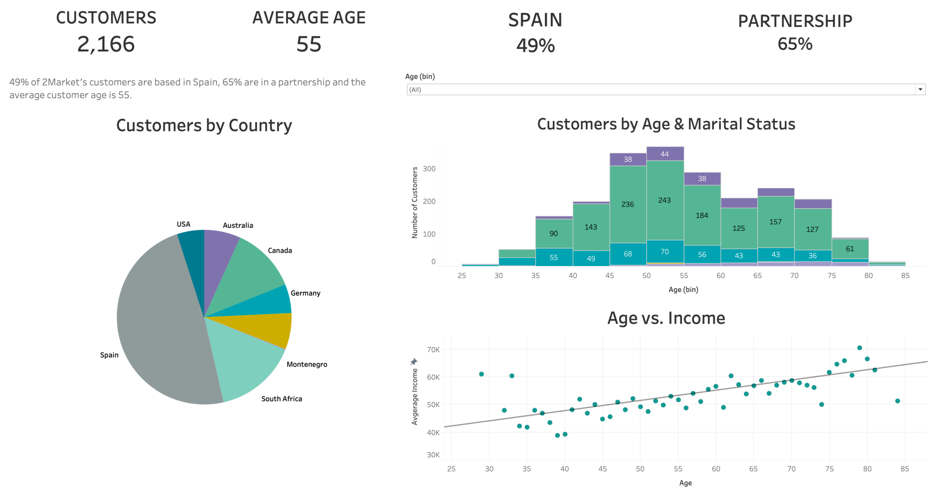

Figure 1: customer demographics dashboard.

For the sales dashboard (Figure 2), sales by country were normalised by calculating average sales per customer rather than total sales to remove population bias resulting from Spain representing nearly 50% of the customer base. The resulting bar chart shows that Canada records the highest average sales, followed by the United States, Germany, and South Africa, although differences between these countries are marginal. Montenegro was excluded from this analysis due to its small sample size of three customers, which significantly distorted the results.

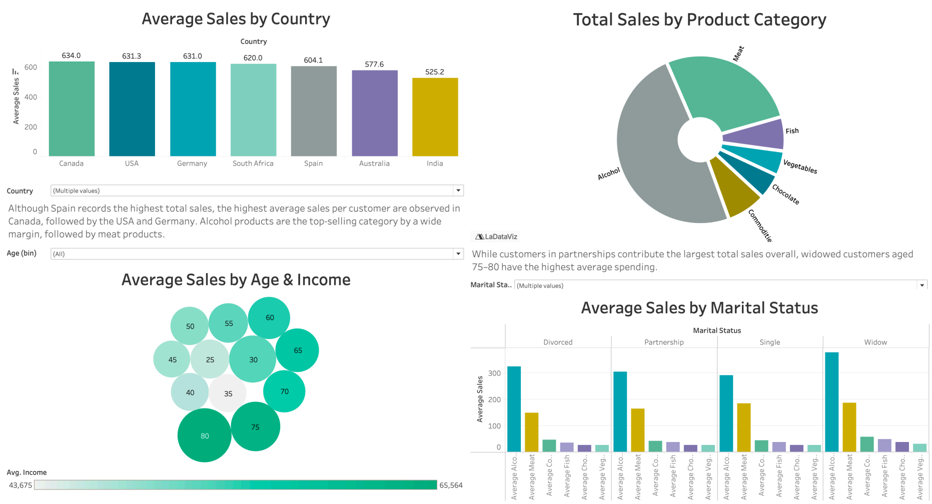

Figure 2: sales trends dashboard.

As more than half of the customers are in a partnership, average sales per customer were also calculated to prevent results being distorted by group size. Across all marital groups, alcohol remains the highest-selling product category, followed by meat. However, the analysis reveals that widowed customers demonstrate the highest average spend, followed by divorced customers, with customers in partnerships ranking third.

The final dashboard illustrates trends in conversion rates: a scatter plot illustrates a positive correlation between total spending and conversion rate. Conversion peaks around the 50-year age group, as shown in the accompanying bar chart. A heatmap further indicates that Canada and Germany demonstrate the highest average conversion rates, particularly via Twitter and email.

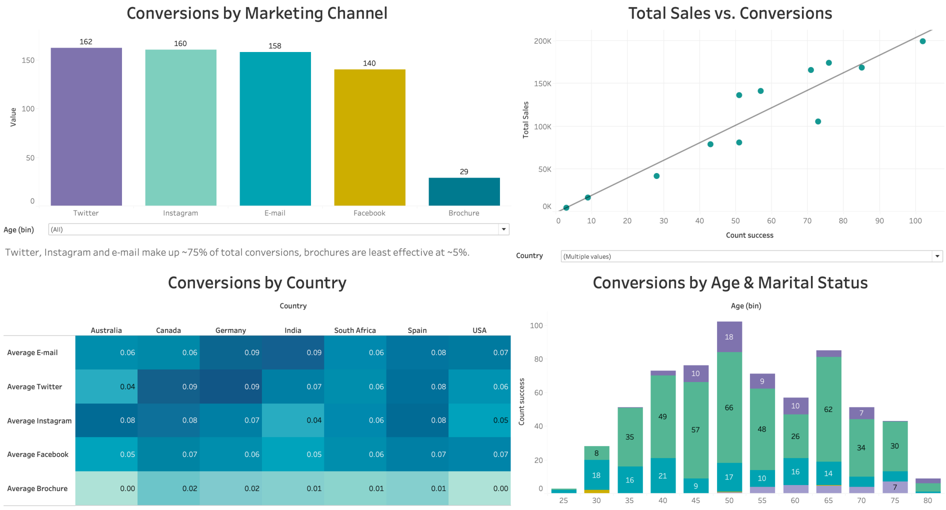

Figure 3: marketing channel success dashboard.

Recommendations

The highest-value customer segment consists of customers aged 75–80, who demonstrate the highest average earnings and spending. This demographic should be prioritised in future campaigns, as they have the potential deliver strong return on marketing investment.

To reduce over-reliance on Spain and mitigate regional risk, market expansion should focus on higher-growth regions, particularly Canada and Germany, where average sales per customer and average conversions rates are strongest. Marketing budgets should be partially reallocated away from lower-performing markets such as India and Australia to support growth in these priority regions.

Digital channels, especially Twitter and Instagram, generate the highest conversion rates and should receive increased investment. Email campaigns should be expanded in Canada, where average customer spending is highest. Brochure advertising consistently underperforms and should be discontinued, with its budget redirected toward higher-performing channels.

While alcohol and meat dominate sales, promotional efforts should increasingly focus on commodities, vegetables, fish, and chocolate products to diversify revenue streams and reduce dependency on a narrow product range.

Further analysis of temporal spending trends would enable evaluation of seasonal campaign effectiveness and customer retention patterns. Additionally, the relationship between the number of children in the household, and spending behaviour warrants further investigation, as it was beyond the scope of this study.

Appendix A

5 Whys Framework

Problem statement:
The marketing team is unsure how to allocate next year's marketing budget to maximise revenue.

Why?
They do not know which marketing channels should receive increased investment.

Why?
They lack clear visibility into which channels generate the highest conversion rates and have the highest sales impact.

Why?
They do not fully understand how different marketing strategies influence customer purchasing behaviour.

Why?
They cannot identify how different customer segments respond to different marketing channels.

Why?
They lack a comprehensive understanding of their customer demographics and behavioural patterns across different locations.

Root cause:
Insufficient insight linking customer demographics, marketing channel performance, and purchasing behaviour.

Appendix B (also available in attached .sql document)

SQL syntax

```
-- create marketing data table:

CREATE TABLE marketing_data(
        "ID" BIGSERIAL PRIMARY KEY,
        "Year_Birth" NUMERIC(4),
        "Age" NUMERIC,
        "Education" VARCHAR(30),
        "Marital_Status" VARCHAR(30),
        "Income" VARCHAR(10),
        "Kidhome" INTEGER,
        "Teenhome" INTEGER,
        "Dt_Customer" VARCHAR(10),
        "Recency" INTEGER,
        "AmtLiq" NUMERIC,
        "AmtVege" NUMERIC,
        "AmtNonVeg" NUMERIC,
        "AmtPes" NUMERIC,
        "AmtChocolates" NUMERIC,
        "AmtComm" NUMERIC,
        "NumDeals" NUMERIC,
        "NumWebBuy" NUMERIC,
        "NumWalkinPur" NUMERIC,
        "NumVisits" NUMERIC,
        "Response" NUMERIC,
        "Complain" NUMERIC,
        "Country" VARCHAR(10),
        "Count_success" NUMERIC);

SELECT * FROM marketing_data;

-- create ad data table:

CREATE TABLE ad_data(
        "ID" BIGSERIAL PRIMARY KEY,
        "Bulkmail_ad" INTEGER,
        "Twitter_ad" INTEGER,
        "Instagram_ad" INTEGER,
        "Facebook_ad" INTEGER,
        "Brochure_ad" INTEGER);
```

```
SELECT * FROM ad_data;

SELECT DISTINCT "Country"
	FROM marketing_data;

-- show number of customers from each country:

SELECT "Country", COUNT("Country") AS number_of_customers
FROM marketing_data
GROUP BY "Country"
ORDER BY number_of_customers DESC;

-- show number of customers from each country and % of total customers for each country:

SELECT
	"Country",
	COUNT(*) AS number_of_customers,
	ROUND(
		COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2
	) AS pct_of_total
FROM marketing_data
GROUP BY "Country"
ORDER BY number_of_customers DESC;


-- show total sales per country, and percentage of total sales:

SELECT
	"Country",
	SUM(
		"AmtLiq"
		+ "AmtVege"
		+ "AmtNonVeg"
		+ "AmtPes"
		+ "AmtChocolates"
		+ "AmtComm"
		) AS total_sales,
		ROUND(
			SUM(
				"AmtLiq"
				+ "AmtVege"
```

```
                                        + "AmtNonVeg"
                                        + "AmtPes"
                                        + "AmtChocolates"
                                        + "AmtComm") * 100.0
                                        / SUM(
                                                  SUM(
                                                            "AmtLiq"
                                                            + "AmtVege"
                                                            + "AmtNonVeg"
                                                            + "AmtPes"
                                                            + "AmtChocolates"
                                                            + "AmtComm"
                                                            )
                                        ) OVER(), 2
                    ) AS pct_of_total
FROM marketing_data
GROUP BY "Country"
ORDER BY total_sales DESC;

-- show total sales per category:

SELECT SUM("AmtLiq") AS liquor_sales,
          SUM("AmtVege") AS vegetable_sales,
          SUM("AmtNonVeg") AS meat_sales,
          SUM("AmtPes") AS fish_sales,
          SUM("AmtChocolates") AS chocolate_sales,
          SUM("AmtComm") AS commodities_sales
FROM marketing_data;

-- show total sales per category per country:

SELECT
          "Country",
          SUM("AmtLiq") AS liquor_sales,
          SUM("AmtVege") AS vegetable_sales,
          SUM("AmtNonVeg") AS meat_sales,
          SUM("AmtPes") AS fish_sales,
          SUM("AmtChocolates") AS chocolate_sales,
          SUM("AmtComm") AS commodities_sales,
          SUM("AmtLiq" + "AmtVege" + "AmtNonVeg" + "AmtPes" + "AmtChocolates" + "AmtComm") AS total_sales
FROM marketing_data
GROUP BY "Country"
```

```
ORDER BY total_sales DESC;

-- create temporary table to show total sales per category per country:

CREATE TEMP TABLE country_sales AS
SELECT
	"Country",
	SUM("AmtLiq") AS liquor_sales,
	SUM("AmtVege") AS vegetable_sales,
	SUM("AmtNonVeg") AS meat_sales,
	SUM("AmtPes") AS fish_sales,
	SUM("AmtChocolates") AS chocolate_sales,
	SUM("AmtComm") AS commodities_sales,
	SUM("AmtLiq" + "AmtVege" + "AmtNonVeg" + "AmtPes" + "AmtChocolates" + "AmtComm") AS total_sales
FROM marketing_data
GROUP BY "Country";

-- show total liquor and meat sales and their percentages, per country:

SELECT
	"Country",
	liquor_sales,
	ROUND(
		liquor_sales * 100.0
		/ SUM(liquor_sales) OVER(), 2
	) AS pct_of_total_liquor_sales,
	meat_sales,
	ROUND(
		meat_sales * 100.0
		/ SUM(meat_sales) OVER(), 2
	) AS pct_of_total_meat_sales,
	total_sales
FROM country_sales
ORDER BY total_sales DESC;

SELECT * FROM marketing_data;

-- show total sales per category per marital status:

SELECT
	"Marital_Status",
	SUM("AmtLiq") AS liquor_sales,
```

```sql
        SUM("AmtVege") AS vegetable_sales,
        SUM("AmtNonVeg") AS meat_sales,
        SUM("AmtPes") AS fish_sales,
        SUM("AmtChocolates") AS chocolate_sales,
        SUM("AmtComm") AS commodities_sales,
        SUM("AmtLiq" + "AmtVege" + "AmtNonVeg" + "AmtPes" + "AmtChocolates" + "AmtComm") AS total_sales
FROM marketing_data
GROUP BY "Marital_Status"
ORDER BY total_sales DESC;

-- total sales per category per kids home:

SELECT
        "Kidhome",
        SUM("AmtLiq") AS liquor_sales,
        SUM("AmtVege") AS vegetable_sales,
        SUM("AmtNonVeg") AS meat_sales,
        SUM("AmtPes") AS fish_sales,
        SUM("AmtChocolates") AS chocolate_sales,
        SUM("AmtComm") AS commodities_sales,
        SUM("AmtLiq" + "AmtVege" + "AmtNonVeg" + "AmtPes" + "AmtChocolates" + "AmtComm") AS total_sales
FROM marketing_data
GROUP BY "Kidhome"
ORDER BY total_sales DESC;

-- total sales per category per teens home:

SELECT
        "Teenhome",
        SUM("AmtLiq") AS liquor_sales,
        SUM("AmtVege") AS vegetable_sales,
        SUM("AmtNonVeg") AS meat_sales,
        SUM("AmtPes") AS fish_sales,
        SUM("AmtChocolates") AS chocolate_sales,
        SUM("AmtComm") AS commodities_sales,
        SUM("AmtLiq" + "AmtVege" + "AmtNonVeg" + "AmtPes" + "AmtChocolates" + "AmtComm") AS total_sales
FROM marketing_data
GROUP BY "Teenhome"
ORDER BY total_sales DESC;
```

```sql
-- join both tables:

CREATE TABLE sales AS
	SELECT
	md."ID",
	"Age",
	"Country",
	"Education",
	"Marital_Status",
	"Income",
	"Kidhome",
	"Teenhome",
	"AmtLiq",
	"AmtVege",
	"AmtNonVeg",
	"AmtPes",
	"AmtChocolates",
	"AmtComm",
	(
		"AmtLiq"
		+ "AmtVege"
		+ "AmtNonVeg"
		+ "AmtPes"
		+ "AmtChocolates"
		+ "AmtComm"
	) AS total_sales,
	ad."Bulkmail_ad",
	"Twitter_ad",
	"Instagram_ad",
	"Facebook_ad",
	"Brochure_ad",
	"Count_success"
	FROM marketing_data md
	JOIN ad_data ad
	USING ("ID");

SELECT * FROM sales;

-- calculate total sales and ads for each customer

SELECT "ID", "Count_success", "total_sales"
FROM sales
```

```sql
ORDER BY "Count_success" DESC;

-- group ads and sales by country:

CREATE TEMP TABLE country_ads AS
SELECT "Country",
        SUM("Bulkmail_ad") AS bulkmail,
        SUM("Twitter_ad") AS twitter,
        SUM("Instagram_ad") AS instagram,
        SUM("Facebook_ad") AS facebook,
        SUM("Brochure_ad") AS brochure,
        SUM("Count_success") AS conversions,
        SUM("total_sales") AS total_sales
FROM sales
GROUP BY "Country"
ORDER BY total_sales DESC;

SELECT * FROM country_ads;

-- calculate total conversions per marketing channel, and their % of total conversions:

SELECT channel,
        total_conversions,
        ROUND(total_conversions * 100.0 / SUM(total_conversions) OVER(), 2
        ) AS pct_of_total
FROM (
SELECT 'bulkmail' AS channel, SUM("bulkmail") AS total_conversions
FROM country_ads
UNION
SELECT 'twitter', SUM("twitter")
FROM country_ads
UNION
SELECT 'instagram', SUM("instagram")
FROM country_ads
UNION
SELECT 'facebook', SUM("facebook")
FROM country_ads
UNION
SELECT 'brochure', SUM("brochure")
FROM country_ads)
ORDER BY total_conversions DESC;
```

```sql
-- group ads and sales by marital status:

SELECT "Marital_Status",
        SUM("Bulkmail_ad") AS bulkmail,
        SUM("Twitter_ad") AS twitter,
        SUM("Instagram_ad") AS instagram,
        SUM("Facebook_ad") AS facebook,
        SUM("Brochure_ad") AS brochure,
        SUM("Count_success") AS conversions,
        SUM("total_sales") AS total_sales
FROM sales
GROUP BY "Marital_Status"
ORDER BY total_sales DESC;

-- calculate sales per product per country and ads per country

SELECT "Country",
        SUM("AmtLiq") AS liquor_sales,
        SUM("AmtVege") AS vegetable_sales,
        SUM("AmtNonVeg") AS meat_sales,
        SUM("AmtPes") AS fish_sales,
        SUM("AmtChocolates") AS chocolate_sales,
        SUM("AmtComm") AS commodities_sales,
        SUM("Bulkmail_ad") AS bulkmail,
        SUM("Twitter_ad") AS twitter,
        SUM("Instagram_ad") AS instagram,
        SUM("Facebook_ad") AS facebook,
        SUM("Brochure_ad") AS brochure,
        SUM("Count_success") AS conversions,
        SUM("total_sales") AS total_sales
FROM sales
GROUP BY "Country"
ORDER BY total_sales DESC;
```

Appendix C

Tables illustrating output of SQL queries

	Marital_Status character varying (30)	liquor_sales numeric	vegetable_sales numeric	meat_sales numeric	fish_sales numeric	chocolate_sales numeric	commodities_sales numeric	total_sales numeric
1	Partnership	423996	35658	228738	51494	37061	60089	837036
2	Single	135244	12468	85070	17826	12471	20185	283264
3	Divorced	73094	6080	33634	7765	5962	10613	137148
4	Widow	27401	2228	13646	3631	2739	4144	53789
5	Unknown	1355	175	725	419	67	492	3233

Table C1: sales per product category and total sales grouped by marital status group.

	Country character varying (10)	liquor_sales numeric	vegetable_sales numeric	meat_sales numeric	fish_sales numeric	chocolate_sales numeric	commodities_sales numeric	total_sales numeric
1	SP	324299	27108	172348	38416	29069	44911	636151
2	SA	103851	8527	56780	13315	8652	14709	205834
3	CA	83893	7678	45855	9963	7592	12096	167077
4	AUS	42424	3680	22204	5534	4105	7023	84970
5	IND	35904	3594	23352	4669	3096	5957	76572
6	GER	36776	2980	20272	4601	2801	5768	73198
7	US	32214	3034	20185	4411	2863	4839	67546
8	ME	1729	8	817	226	122	220	3122

Table C2: sales per product category and total sales grouped by country.